1. Applying Mathematical Knowledge to Solve Real-World Problems

(Corresponds to "employ.pdf" emphasis on practical mathematical literacy)

- Question Content: The problem presents a revolving door with three wings dividing a circular space into three equal sectors. Students must calculate the angle between two adjacent wings.
- Relevance Analysis:
  - This requires recognizing that a full circle is 360° and dividing it equally among three sectors (360° ÷ 3 = 120°).
  - The task directly applies geometric knowledge (angles in a circle) to a real-world architectural scenario, aligning with "employ.pdf"'s focus on using mathematics to address practical, context-based challenges.

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2. Interpreting Diagrams and Spatial Reasoning

(Corresponds to "employ.pdf" emphasis on visual and spatial literacy)

- Question Content: Students must interpret a top-down plan of the revolving door to understand how the wings partition the circular space.
- Relevance Analysis:
  - Success depends on translating a 2D diagram into a mental model of the physical structure.
  - This reflects "employ.pdf"'s priority on interpreting visual representations and applying spatial reasoning to solve problems.

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3. Demonstrating Foundational Mathematical Competence

(Corresponds to "employ.pdf" emphasis on core mathematical skills)

- Question Content: The problem tests basic understanding of angles and division of a circle.
- Relevance Analysis:
  - While simple, the question assesses essential geometric knowledge required for more complex real-world applications.
  - This aligns with "employ.pdf"'s goal of ensuring individuals possess fundamental mathematical competencies necessary for everyday problem-solving.

1. 将数学知识应用于解决现实世界的问题

（对应《employ.pdf》强调的实用数学素养）

- 题目内容：题目描述了一扇由三片翼板组成的旋转门，将一个圆形空间均分为三个相等的扇形区域。学生需要计算相邻两片翼板之间的夹角。
- 相关性分析：
  - 学生需知道一个完整的圆为360°，并将其平均分成三份（360° ÷ 3 = 120°）。
  - 这一任务直接将几何知识（圆的角度）应用于现实中的建筑场景，与《employ.pdf》所强调的“运用数学解决基于真实情境的实际问题”高度契合。

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2. 解读图示与空间推理能力

（对应《employ.pdf》对视觉和空间素养的重视）

- 题目内容：学生必须解读旋转门的俯视图，理解翼板如何将圆形区域划分。
- 相关性分析：
  - 解题依赖于将二维图示转化为对物理结构的心理模型。
  - 这体现了《employ.pdf》所重视的“解读视觉信息并运用空间推理解决问题”的能力。

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3. 展现基础数学能力

（对应《employ.pdf》对核心数学技能的要求）

- 题目内容：题目考查了对角度及圆的分割等基本几何概念的理解。
- 相关性分析：
  - 虽然题目简单，但它评估了解决更复杂现实问题所需的基本几何知识。
  - 这符合《employ.pdf》的目标，即确保个体具备“日常问题解决所必需的基础数学能力”。